IA -3

Take home assignment

Last day for submission: 03/11/2025

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1. A 9375 MHz uniform plane wave is propagating in polystyrene. If the amplitude of electric field intensity is 20 V/m and the material is assumed to be lossless, find
   a) β
   b) λ in polystyrene
   c) Velocity
   d) η
   e) Amplitude of magnetic field intensity

2. An electromagnetic wave in free space has a wavelength of 0.20 m. When this same wave enters a perfect dielectric, the wavelength changes to 0.09 m. Assuming that $\mu_r = 1$, determine ϵ_r and the wave velocity in the dielectric.

3. The amplitude of **E** just inside a liquid is 10.0 V/m and the constants are $\mu_r = 1$, $\epsilon_r = 20$, and $\sigma = 0.50$ S/m. Determine the amplitude of **E** at a distance of 10 cm inside the medium for frequencies of (*a*) 5 MHz, (*b*) 50 MHz, and (*c*) 500 MHz.

4. The electric field of a TV broadcast signal progagating in air is given by

$$\mathbf{E}(z,t) = 0.2\cos(\omega t - 6.5z)\mathbf{a}_x \text{ V/m}$$

   (a) Determine the wave frequency ω and the wavelength λ.
   (b) Sketch $\mathbf{E}_x$ as a function of t at $z = 0$ and $z = \lambda/2$.
   (c) Find the corresponding $\mathbf{H}(z, t)$.

5. A manufacturer produces a ferrite material with $\mu = 750\mu_o$, $\varepsilon = 5\varepsilon_o$, and $\sigma = 10^{-6}$ S/m at 10 MHz.
   (a) Would you classify the material as lossless, lossy, or conducting?
   (b) Calculate β and λ.
   (c) Determine the phase difference between two points separated by 2 m.
   (d) Find the intrinsic impedance.

Write your Register Number and Sign on each page, scan and upload the assignment before the due date.